



UNIVERSITÀ
DI TORINO



MSSC2025
TORINO 8-12 SEP 2025

TODAY'S PROGRAM

TUESDAY 09TH

AFTERNOON ADVANCED TUTORIALS

Copernico Room

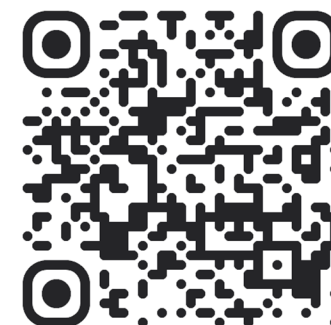
- 14:30 - 15:15 **Equation-of-State and Pressure** (Filippo Bodo)
- 15:15 - 16:00 **Elastic Properties** (Filippo Bodo)
- 16:00 - 16:30 **Ice Cream Break**
- 16:30 - 18:00 **Electron Transport Properties** (Eleonora Ascrizzi)

AFTERNOON BASIC TUTORIALS

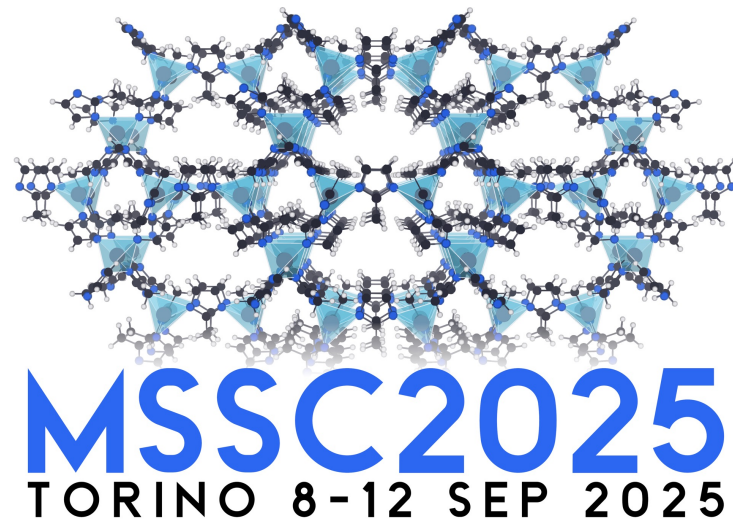
von Neumann Room

- 14:30 - 15:15 **Geometry Optimization and Equation-of-State** (Giuseppe Mallia)
- 15:15 - 16:00 **Modelling of 2D Systems** (Giuseppe Mallia)
- 16:00 - 16:30 **Ice Cream Break**
- 16:30 - 18:00 **Harmonic Frequencies and Spectra** (Davide Mitoli)

View the full
timetable online



Ab Initio Modelling in Solid State Chemistry

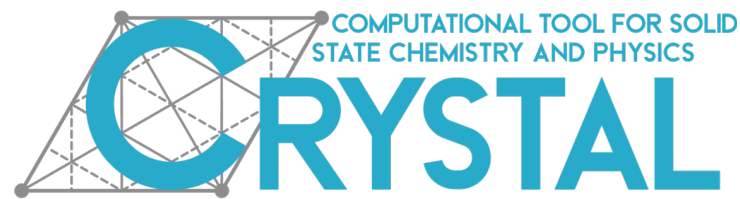


Harmonic Frequencies and Spectra

Davide Mitoli
(davide.mitoli@unito.it)



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Lattice Dynamics

- ✓ Atoms are never static → lattice dynamics explains thermal effects, transport properties, phase transitions, vibrational spectra, thermodynamic properties, ...
- ✓ Ab initio simulations of atomic vibrations requires solving the nuclear Schrödinger equation → within the HA this means:
 1. Computing the Hessian of the electronic (DFT) energy
 2. Diagonalize the mass-weighted Hessian
- ✓ More on this topic tomorrow morning, let's focus on a hands-on approach today!

Tutorial Outline

- ✓ Computing harmonic vibrational frequencies with CRYSTAL
 - ✓ The theoretical minimum
 - ✓ Input/Output tour
- ✓ IR and Raman intensities
 - ✓ Berry-Phase (numerical)
 - ✓ CPHF/KS (analytical)
- ✓ Plotting Vibrational Spectra:
 - ✓ Optional keywords
 - ✓ Tools (CRYSLOT, CRYSTALClear,...)

Vibrational Frequencies in CRYSTAL (I)

- ✓ Vibrational frequencies calculation is performed at the Γ point within the harmonic approximation.
- ✓ The dynamical matrix, H_{ij} (i.e. the matrix of the second-derivatives), is computed by numerical evaluation of the first-derivatives

$$H_{ij} = \left(\frac{\partial g_i}{\partial u_j} \right)_{\text{eq}} \approx \frac{g_i(0, \dots, u_j, 0, \dots)}{u_j}$$

of the analytical atomic gradients

$$g_i = \left(\frac{\partial E}{\partial u_i} \right)_{\text{eq}} \quad \{u_i\} \text{ are displacement vectors}$$

Vibrational Frequencies in CRYSTAL (II)

- ✓ The mass-weighted Hessian is calculated:

$$W_{\alpha i, \beta j}(\mathbf{k} = \mathbf{0}) = \frac{H_{ij}}{\sqrt{M_{\alpha} M_{\beta}}}$$

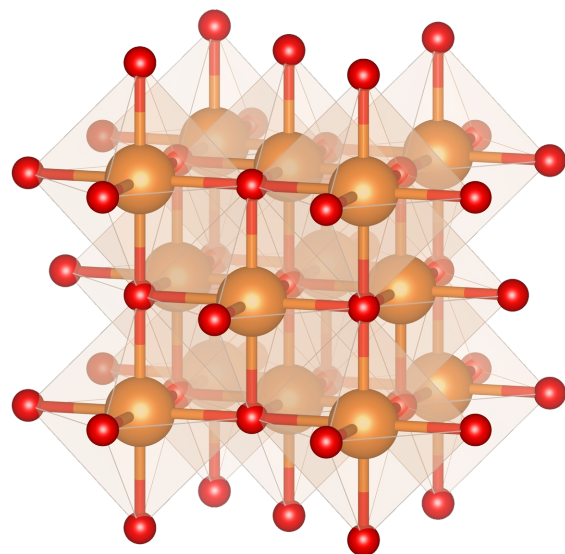
Where M_{α} and M_{β} are the masses of the atoms α and β associated with the i -th and j -th coordinates, respectively.

- ✓ The mass-weighted Hessian is diagonalized:
 - Eigenvalues $\rightarrow$ frequencies
 - Eigenvectors $\rightarrow$ normal modes

Vibrational Frequencies in CRYSTAL (III)

- ✓ Special properties of the Γ point ($\mathbf{k} = \mathbf{0}$):
 - The Hessian is simple to calculate
 - 3 modes have zero frequency (i.e., acoustic modes that correspond to translations in Γ)
 - The Hessian possesses the point symmetry of the crystal
 - Γ -point vibrational modes give rise to IR and Raman spectra
- ✓ A supercell approach can be adopted to include more points of the first Brillouin zone
 - Phonon dispersion (`SCELPHONO` keyword)
 - Relevant for accurate thermodynamic properties
- ✓ The point group symmetry of the system is fully exploited to reduce the number of calculations (SCF+gradient) to be considered. On each numerical step, the residual symmetry is preserved during the SCF process and the gradients calculation.

Input for Frequency Calculation



MgO - PBE/pob-DVZVP-rev2

CRYSTAL

0 0 0

225

4.26844350

2

12 0. 0. 0.

8 0.5 0.5 0.5

FREQCALC

END

BASISSET

POB-DZVP-REV2

DFT

PBE

END

SHRINK

8 8

END

Geometry Block

Basis set

DFT functional

Shrinking factor

Infrared Intensities in CRYSTAL

- ✓ The (harmonic) IR intensity I_i^{IR} of the i -th mode is defined as

$$I_i^{\text{IR}} \propto d_i \left(\frac{\partial \mu}{\partial Q_i} \right)^2 \quad \text{Born dynamical charges}$$

i.e., it is proportional to the first-derivative of the electric dipole moment μ with respect to the normal mode Q_i times the degeneracy d_i of the i -th mode.

- ✓ Born dynamical charges can be computed in 3 different ways:
 - ✓ Berry phase (default)
 - ✓ Wannier functions
 - ✓ Coupled-perturbed Hartree-Fock/Kohn-Sham

Input for IR Intensities

FREQCALC block	FREQCALC	
	<i>RESTART</i>	} Restart from previous calculation
	INTENS	
	<i>INTCPHF</i>	} CPHF/KS analytical approach
	<i>END</i>	
	<i>IRSPEC</i>	} Simulate IR spectrum and store it in IRSPEC.DAT
	<i>DAMPFAC</i>	
	<i>4.0</i>	} Full width at half maximum (FWHM), default is 8.0
	<i>RANGE</i>	
	<i>1200 1500</i>	} Frequency range, default is [0.0, $\nu_{\max} \times 1.3$]
	END	

Keywords in *italic* are optional

FREQCALC Output (I)

[illegible]

FORCE CONSTANT MATRIX - NUMERICAL ESTIMATE

[illegible]

MAX ABS (DGRAD) : MAXIMUM ABSOLUTE GRADIENT DIFFERENCE WITH RESPECT TO

THE CENTRAL POINT

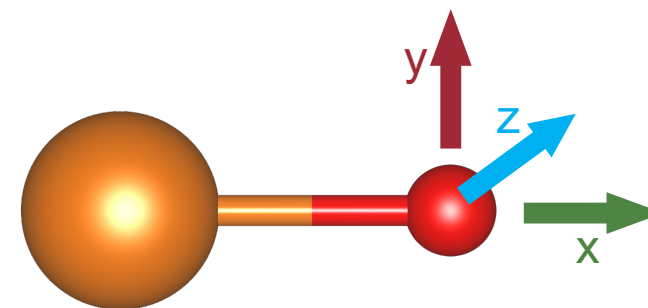
DE: ENERGY DIFFERENCE WITH RESPECT TO THE CENTRAL POINT

(DE IS EXPECTED TO BE POSITIVE FOR ALL DISPLACEMENTS)

ATOM	MAX ABS (DGRAD)	TOTAL ENERGY (AU)	N.CYC	DE	SYM
CENTRAL POINT		-2.751735599829E+02	8	0.0000E+00	48
2 O DX	2.9289E-04	-2.751735591513E+02	4	8.3165E-07	8
2 O DY		GENERATED FROM A PREVIOUS LINE			
2 O DZ		GENERATED FROM A PREVIOUS LINE			

The mass-weighted Hessian H_{ij} is computed by numerical evaluation of first-derivatives of the analytical atomic gradients

$$H_{ij} = \left(\frac{\partial g_i}{\partial u_j} \right)_0 \approx \frac{g_i(0, \dots, u_j, 0, \dots)}{u_j}$$



FREQCALC Output (II)

Upon diagonalization of the mass-weighted Hessian...

CONVERSION FACTORS FOR FREQUENCIES:

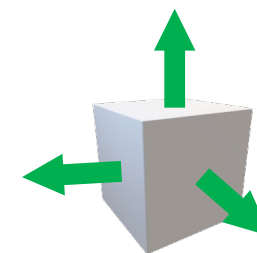
```
1 CM**(-1) = 0.4556335E-05 HARTREE
```

```
1 THZ      =      0.3335641E+02  CM** (-1)
```

[illegible]

MODES		EIGV	FREQUENCIES		IRREP	IR	INTENS	RAMAN
		(HARTREE**2)	(CM**-1)	(THZ)	(KM/MOL)			
1-	3	0.4758E-21	0.0000	0.0000	(Fu)	A (	0.00)	I
4-	6	0.2954E-05	377.1845	11.3077	(Fu)	A (	1202.18)	I

rigid translations of
the whole crystal



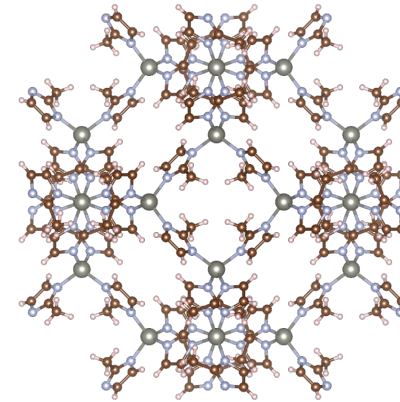
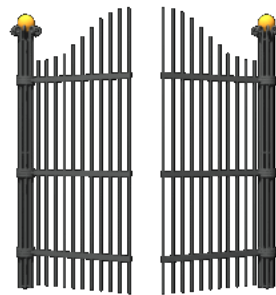
FREQCALC Output (III)

NORMAL MODES NORMALIZED TO CLASSICAL AMPLITUDES (IN BOHR)

FREQ (CM** -1)			0.00	0.00	0.00	377.18	377.18	377.18
AT. 1 MG	X	0.1582	0.0000	0.0000	0.0000	-0.0730	0.0000	
	Y	0.0000	0.1582	0.0000	-0.0730	0.0000	0.0000	
	Z	0.0000	0.0000	0.1582	0.0000	0.0000	-0.0730	
AT. 2 O	X	0.1582	0.0000	0.0000	0.0000	0.1094	0.0000	
	Y	0.0000	0.1582	0.0000	0.1094	0.0000	0.0000	
	Z	0.0000	0.0000	0.1582	0.0000	0.0000	0.1094	

Exercise

1. Perform a calculation of the IR spectrum of MgO. Make use of the Berry-phase approach.
2. Perform the same calculation by making use of the CPKS approach (slow).
3. Use **CRYSPLOT** to visualize lattice vibrations and IR spectrum of MgO (quite boring...).
4. Use **CRYSPLOT** to visualize lattice vibrations and IR spectrum of ZIF-8 (way more interesting!). Can you identify the frequencies corresponding to the “gate-opening” modes?



Raman Intensities in CRYSTAL

- ✓ The Raman intensity for an oriented single-crystal (let's consider the ij directions) associated to the mode with frequency ω_k is:

$$I_{ij}^{\text{Raman}} \propto C (\alpha_{ij}^k)^2$$

where the α_{ij}^k element of the Raman tensor is given by:

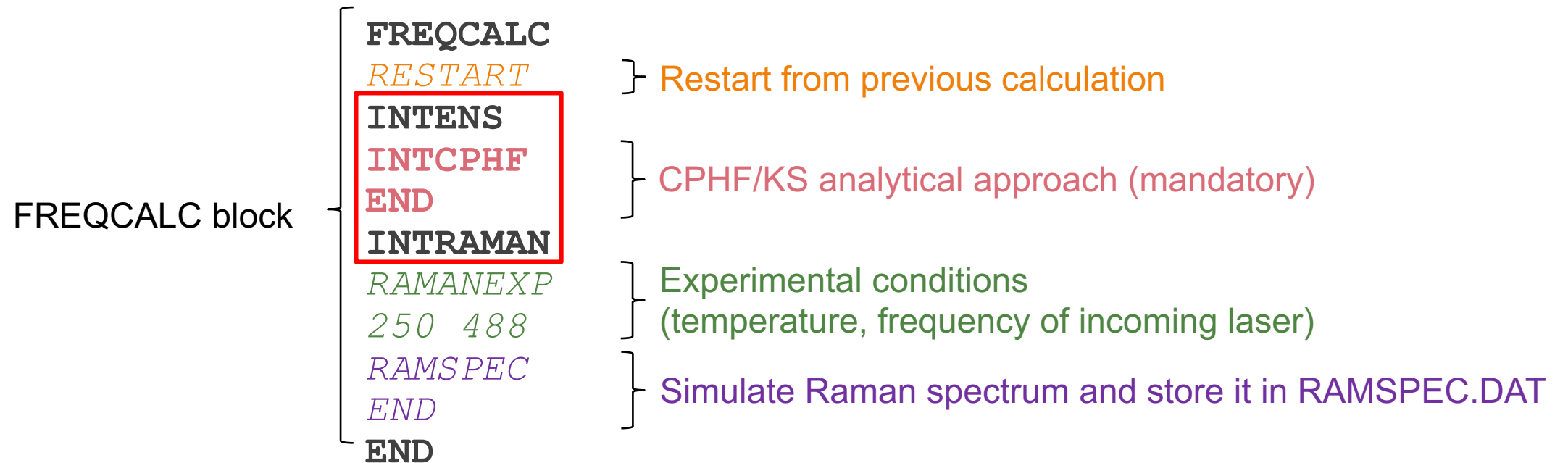
$$\alpha_{ij}^k = \frac{\partial^3 E}{\partial Q_k \partial \mathcal{E}_i \partial \mathcal{E}_j}$$

where $\vec{\mathcal{E}}$ is the external electric field.

The pre-factor C depends on the laser frequency ω_L and on temperature T as follows

$$C \propto \frac{1 + n(\omega_k)}{30\omega_k} (\omega_L - \omega_k)^4 \quad \text{with} \quad 1 + n(\omega_k) = \left[1 - \exp\left(-\frac{\hbar\omega_k}{K_B T}\right) \right]^{-1}$$

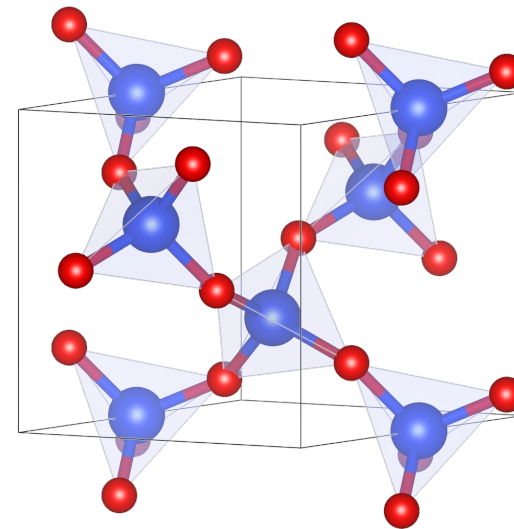
Input for Raman Intensities



Keywords in *italic* are optional

Exercise

1. Starting from the α -quartz input `qua_hf_2d_f-raman_295K_488nm.d12`, insert the appropriate keywords to calculate the RAMAN intensities at 10 K and 400 K, using an incident radiation wavelength of 550 nm (use the **RAMANREA** keyword).
2. Simulate the corresponding Raman spectra (set the FWHM to 3).
3. Compare the two spectra using  .



Exercise

1. Calculate the harmonic frequencies and IR intensities of calcite (RESTART).
2. Which are the irreducible representations? Which modes are IR active? And Raman active?
3. Analyze the normal modes by using . Which modes correspond to rigid shifts of the Ca lattice?

